

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Hosted by: Moses Lake High School

GENERAL INSTRUCTIONS applying to all tests:

- *Good sportsmanship is expected throughout the competition by all involved (competitors and observers). Display of poor sportsmanship will result in disqualification.*
- *Competitors may not use calculators or any other devices or aids on any portion of this contest.*
- *Unless stated otherwise:*
 - *All answers are integers, and any non-integer answers will be "coded" as integers.*
 - *For 5th grade and up, all fractions and ratios must be reduced to simplest form, all radicals must be simplified, and all denominators must be rationalized.*
 - *Do not round or approximate answers. Leave answers in terms of π or other irrational quantities (e.g., $\sqrt{2}$), where applicable.*
- *Units are not necessary as part of your answer, However, if you choose to use units, they must be correct.*
- *Record all answers on the colored cover sheets in the answer column only.*
- ***Be sure that the student name, school, team number, etc. has been filled out at the top of each answer sheet.***
- *Tests will be scored as a 0 if answers are not recorded correctly on the answer sheets.*
- *Blank answer sheets and answer sheets with no name will be scored as a 0.*

FINAL SCORES AND AWARDS

Individual awards are determined by both the Mental Math and Individual Test scores. Individual ties are broken based on the following, in this order: total scaled individual points, total number of correct answers on the Individual Test, Mental Math raw score, number of correct answers from Individual Test #31-45, number of correct answers from Individual Test #16-30, highest numbered question answered correctly on the Individual Test working backwards from #45.

Team (School) awards are based on the highest score from amongst each of the school's "teams of 4 students" in each event and is calculated as $2 \cdot (\text{Sum of highest 3 Mental Math scores}) + 2 \cdot (\text{Multiple Choice}) + 6 \cdot (\text{Team}) + 1 \cdot (\text{Triple Jump}) + 1 \cdot (\text{College Bowl})$, for approximate weights of 25%, 20%, 30%, 15% and 10% respectively. Team ties are broken based on highest event score in order of the events, starting with Mental Math.

MENTAL MATH TEST - 30 sec./quest., 8 problems, ~8%/25% of individ./team scores

The proctor will read each question twice. You may not do any writing or talking while arriving at a solution. Record only your answer on your answer sheet. You may not change, cross out, erase, or write over an answer once you have written it down. The maximum wait time is 30 seconds after completion of the second reading of the question. Correct answers receive 1 point.

INDIVIDUAL TEST - 35 minutes, 45 problems, ~92% of individual score

When you are prompted to begin, tear off the colored answer sheet and begin testing. No talking during this individual test. You will be given a 5 minute time warning. Correct answers receive 2 points for problems 1-30 and 3 points for 31-45 (in the scaled score).

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Final Score (out of 8)

Room #

School Name

Student Name

Team #

Mental Math - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

STUDENT: DO NOT WRITE IN SHADED REGIONS (or anywhere else, other than the answer box)

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
7 th /8 th grade		TOTAL:	

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Key

Mental Math Contest – Answer Key

30 seconds per question – ~25% of team score & ~8% of individual score

SCORERS — Write-overs, Cross-outs, and Erasures Must be Marked Incorrect (0)
Bracketed items [...] in the answer key are optional.

7th/8th grade

Answer		
1	50	What is twenty-five times four minus twenty-five times two?
2	24	Evaluate ten X minus X squared if X equals four.
3	30 [inches]	The lengths of the two legs of a right triangle are five and twelve inches. What is the number of inches in the perimeter of the triangle?
4	31 [nickels]	Eho has one dollar and fifty-seven cents in his pocket. What is the largest number of nickels he could have in his pocket?
5	3520 [feet]	There are five thousand two hundred and eighty feet in a mile. How many feet are in two-thirds of a mile?
6	45	A sequence begins with the numbers three, six, ten, and fifteen. What is the eighth term in this sequence?
7	211	What is the largest three-digit prime number whose digits add up to four?
8	[A + B =] 43	A standard deck of cards has all its face cards and aces removed. As a reduced common fraction, the probability that two cards, randomly selected without replacement, are both red and divisible by three is A over B. What is A plus B?

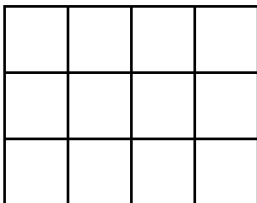
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Individual Contest

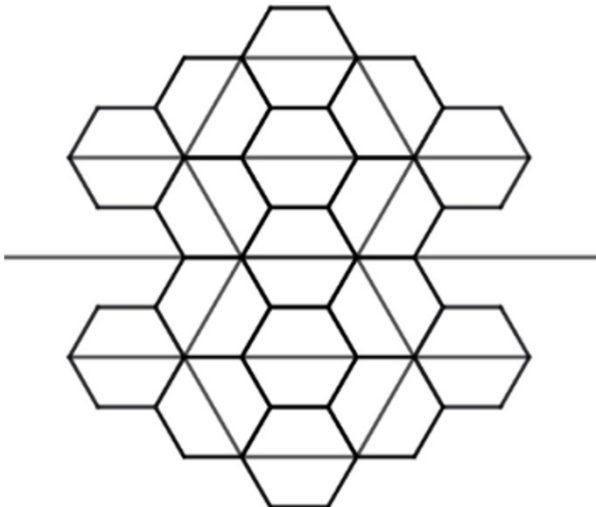
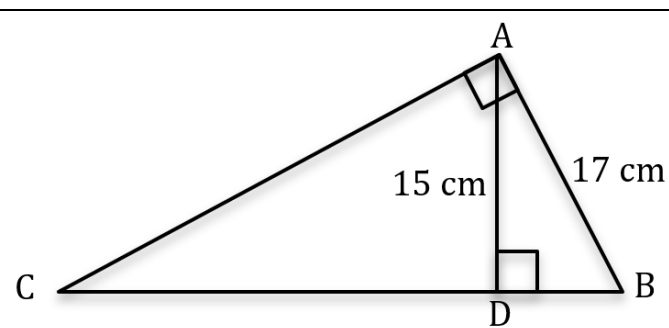
Record all answers on the colored cover sheet. 35 minutes, 45 problems, ~92% of individual score.

No talking during this individual test. A 5-minute time warning will be given.

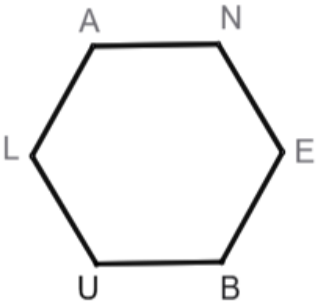
	Questions 1-30: 2 points each	
1	What is 18% of 650?	
2	Evaluate: $(\sqrt{144})^2$	
3	How many square inches are in the area of a rectangle whose length is 23 inches and whose width is 17 inches?	
4	Solve for x: $7x + 61 = 173$	
5	As a simplified fraction, the ratio $930/217 = A/B$. What is $A + B$?	
6	How many squares of any size are in the figure? The figure is made up of unit squares combined to form a rectangle.	
7	What is the mean of the following set of numbers? Your answer should be given as an integer. 15.3, 13.7, 10.4, 18.1, 12.5	
8	How many inches are in $4\frac{2}{3}$ yards?	
9	Cesar reads 9 pages of his book every 7 minutes. At this rate, how long in minutes will it take him to complete his 324-page book?	
10	As a percentage, what is the probability that a number randomly selected from the integers 1 through 20 is a multiple of 7?	
11	Evaluate the expression $\frac{7a}{6} + \frac{15b}{8}$, if $a = 18$ and $b = 32$.	
12	What is the smallest positive 5-digit integer in which three of the digits are distinct odd integers and the other two digits are distinct even integers?	
13	What is the 20 th term in an arithmetic sequence whose 3 rd term is 16 and whose 7 th term is 48?	
14	In the following addition problem, each letter represents a digit from 0 to 9, and $B < A < C$. What is the largest possible three-digit integer CBA that can be formed? $A + B = C$	
Continued on next page.		

15	How many different four-digit integers can be formed by arranging the four digits in the number 2026? Note: a four-digit integer cannot have 0 in the thousands place.
16	A rectangle has an area of 3.7 square meters. How many square centimeters are in the area of the rectangle?
17	A data set consists of 6 distinct positive integers, has a range of 17 and a median of 9. What is the largest possible integer in the set?
18	It takes Nancy 3 hours to paint her room. It would take her little sister 7 hours to paint the same room. If they had worked together, in minutes, how long would it have taken them to paint the room?
19	The prime factorization of 680 can be written in the form $a^3 \cdot b \cdot c$, where a , b , and c are prime numbers. What is $a + b + c$?
20	Let A , B , C , and D , each represent a different positive even single-digit integer. What is the largest possible value of $\frac{A}{B} \cdot \frac{C}{D}$?
21	Exactly one of the following statements is false, and the rest are true. Which statement is false? Answer 1, 2, 3, or 4. 1. Statement 2 is true. 2. Statement 3 is false. 3. Statement 4 is false. 4. Statement 1 is true.
22	Evaluate: $\frac{\frac{2}{3}}{\frac{5}{7}} \cdot \frac{1}{6} + \frac{83}{45}$
23	If $a \Omega b = (2 - a)(7 - b)$, then evaluate $12 \Omega (18 \Omega 4)$.
24	Rafa bought a car for \$25000. Later, he sold it for 10% more than he paid for it. Using all the money he got from selling the car, he bought a new, more expensive car. He then sold the second car for an amount, x , that is 20% less than he paid for it. What percent of the original \$25000 purchase price is x ?
25	Maria walks $\frac{1}{3}$ of a mile in $\frac{1}{7}$ of an hour. At this average rate, how far in miles would she walk in $2\frac{1}{7}$ hours?
26	How many ways are there to arrange the letters in the word ALIEN, if the vowels must always be in alphabetical order (not necessarily consecutive) from left to right?
27	The first 6 terms of an infinite sequence are 1, 2, 2, 3, 3, 3, . . . What is the 100 th term in this sequence?
28	A square has an area of 81 cm^2 . How many circles of radius 2 cm can fit completely inside the square without overlapping?
29	All face cards and aces are removed from a standard deck of cards. As a reduced common fraction, the probability that a hand of three cards drawn from the remaining cards consists of all even cards is A/B . What is $A + B$?
30	What is the smallest positive integer n , for which each of the expressions $\frac{n+2}{3}$, $\frac{3n+1}{4}$, and $\frac{2n+7}{5}$ represents an integer?
Continued on next page.	

Challenge Questions: 3 points each

31	Athena, Joy, and Min simultaneously begin jogging in the same direction from the same point on a circular track. Joy's average speed is 5% slower than Athena's average speed. Athena passes Min for the first time at the moment when Athena completes her sixth lap. What percent faster is Joy's average speed than Min's average speed?
32	Each half of this figure is composed of 4 red trapezoids, 6 blue trapezoids and 7 white trapezoids. When the upper half is folded down over the centerline, 2 pairs of red trapezoids coincide, as do 4 pairs of blue trapezoids. There are 2 blue-white pairs. How many white pairs coincide? 
33	A jar contains 5 red, 4 blue, and 3 green marbles. Three marbles are randomly removed from the jar. As a reduced common fraction, the probability that exactly two of the three marbles are of the same color is A/B . What is $A + B$?
34	In $\triangle ABC$, $\overline{AD} \perp \overline{BC}$, $m\angle BAC = 90^\circ$, $AB = 17$ cm, and $AD = 15$ cm. As a reduced common fraction, the ratio of $BD/AC = P/Q$. What is $P + Q$? 
35	On a coordinate plane, $\triangle ABC$ has vertices $A(0, -10)$, $B(11, -3)$, and $C(0, 5)$. Imagine this triangle is rotated 360° about the y-axis. The volume of the 3-dimensional space through which the triangle rotates can be expressed as $Q\pi$ cubic units. What is the value of Q ?
36	How many positive 4-digit integers have no repeating digits, no digit that is a 2, and have digits whose sum is 24?
37	A number, n , is 2 more than 3 times its reciprocal. What is the sum of all values of n ?
38	What is the maximum value of n , such that $9n$ is a factor of $250!$ (250 factorial)?
39	The median of the following list of 4052 numbers is $A.B$, where A is a 4-digit integer and B is a 1-digit integer. What is $A + B$? $1, 2, 3, \dots, 2026, 1^2, 2^2, 3^2, \dots, 2026^2$

Continued on next page.

40	The lengths of the three sides of a right triangle form a geometric sequence. The ratio of the length of the hypotenuse to the length of the short leg can be expressed as $\frac{R+\sqrt{S}}{T}$, where R , S , and T are each single-digit integers. What is $R + S + T$?
41	Sides $\overline{LA}$ and $\overline{EN}$ in a regular hexagon $NEBULA$, shown here, are extended until they intersect at a point F. If the sides of the hexagon have length 12 units, then the length of $\overline{FB}$ is $P\sqrt{Q}$ units, where P and Q are both integers, and Q has no factors that are perfect squares, other than 1. What is $P + Q$? 
42	Two distinct geometric sequences consist of three integers. One is a_1, a_2, a_3 , and the other is b_1, b_2, b_3 , both sets of three integers have a mean of 21, and $a_1 < b_1$. What is the value of $a_3 - b_3$?
43	What is the sum of all prime numbers less than 40 that are the largest value in a Pythagorean Triple? An example is 5, since it is prime and is the largest value in the 3:4:5 Pythagorean Triple.
44	The solutions $x = u$ and $x = v$ of the quadratic equation $rx^2 + sx + t = 0$ are reciprocals of the solutions of the quadratic equation $(2 + a)x^2 + 5x + (2 - a) = 0$ for some integer a . If the GCF of r, s and t is 1, what is the value of $r + s + t$?
45	For years, Mr. Kangas has had a tradition of giving his students a list of books from which they had to choose two to read during spring break. This year, Mr. Kangas added n book(s) to the list. If there are now 84 more ways that two books can be selected, what is the sum of all possible values of n ?

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KEY

7th/8th grade Individual Contest - Answer Key

SCORERS: Bracketed [...] items in answer key are optional. Just mark the score as 0 or 1 and add up those values to reflect total correct.

First Scorer - use the right-hand columns so 2nd scorer can do a blind scoring.

	Answer
1	117
2	144
3	391 [in ²]
4	[x =] 16
5	[A + B =] 37
6	20 [squares]
7	14 [= mean]
8	168 [inches]
9	252 [minutes]
10	10 [%]
11	81
12	10235
13	152
14	[CBA =] 945
15	9 [integers]

	Answer
16	37000 [cm ²]
17	23
18	126 [minutes]
19	[a + b + c] = 24
20	6
21	3 [= statement no.]
22	2
23	-550
24	[x =] 88 [%]
25	5 [miles]
26	20 [ways]
27	14 [= 100 th term]
28	4 [circles]
29	[A + B =] 138
30	[n =] 71

	Answer
31	14 [%]
32	5 [white pairs]
33	[A + B =] 73
34	[P + Q =] 319
35	[Q =] 605
36	162 [integers]
37	2
38	[n =] 61
39	[A + B =] 1987
40	[R + S + T =] 8
41	[P + Q =] 19
42	[a ₃ - b ₃ =] 12
43	101
44	[r + s + t =] 9
45	[n =] 19

"Math Is Cool" Masters — 2025-26

Total Correct (all columns)

Room #

SCHOOL NAME

STUDENT NAME

Team #

7th/8th Individual Contest - Score Sheet

STUDENTS: DO NOT WRITE IN SHADED REGIONS

	Answer	1 or 0	1 or 0
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
1-15 TOTAL:			

	Answer	1 or 0	1 or 0
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
16-30 TOTAL:			

	Answer	1 or 0	1 or 0
31			
32			
33			
34			
35			
36			
37			
38			
39			
40			
41			
42			
43			
44			
45			
31-45 TOTAL:			

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7th/8th grade — April 25, 2026

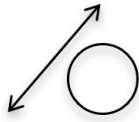
Team Multiple Choice Contest

	USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #4. If $a \Delta b = 3ab$, then $2 \Delta 4 = 3 \cdot 2 \cdot 4 = 24$
1	If $7 \square 11 = 72$, then which of the following rules could define $a \square b$? A) $a \square b = 6a + 3b$ B) $a \square b = 12a - b$ C) $a \square b = b^2 - a^2$ D) $a \square b = a^2 + b^2 - 100$ E) Answer not given.
2	If $a \odot b = a(a + b)/2$, then what is $9 \odot 13$? A) 78 B) 99 C) 149 D) 198 E) Answer not given.
3	Using the functions as defined above, what is the value of: $8 \Delta (6 \square (4 \odot 2))$? A) 864 B) 1744 C) 2592 D) 12960 E) Answer not given.
4	Let a be the arithmetic sequence $a_1, a_2, a_3, \dots, a_n, \dots = 3, 6, 9, \dots$ and let b be the arithmetic sequence $b_1, b_2, b_3, \dots, b_n, \dots = 15, 13, 11, \dots$ and let $f(x) = a_x \boxminus b_x = 8a_x - 5b_x$. For example, $f(1) = a_1 \boxminus b_1 = 8a_1 - 5b_1 = 8 \cdot 3 - 5 \cdot 15 = 24 - 75 = -51$. Which of the following is $f(x)$ in terms of x? A) $f(x) = -1.5x^2 + 38.5x - 88$ B) $f(x) = -0.5x^2 + 35.5x - 86$ C) $f(x) = 33x - 84$ D) $f(x) = 34x - 85$ E) Answer not given.

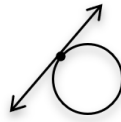
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USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #5 THROUGH #7.

The first set of figures shown demonstrate the different number of points of intersection possible for a line and a circle.



0 points of intersection

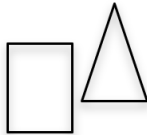


1 point of intersection

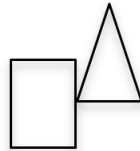


2 points of intersection

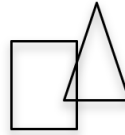
The next set of figures shown demonstrate the different number of points of intersection possible for two or more shapes with straight edges. Note: in the 4th and 5th figures below, one side of the triangle lies directly on top or partly on top of one of the sides of the rectangle.



0 points of intersection



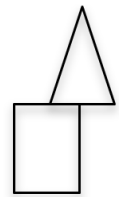
1 point of intersection



2 points of intersection



infinite points of intersection



infinite points of intersection

5 What is the maximum possible number of points of intersection for three lines and a circle? Note: points of intersection can be where one or more lines and the circle intersect, where two or more of the lines intersect each other, or where two or more of the lines and the circle intersect. Also, no line lies on top of any other line.

- A) 5 B) 6 C) 7 D) 8 E) Answer not given.

6 What is the maximum possible number of points of intersection for a square and a regular pentagon whose sides are the same length as the sides of the square? Note: no side of the square lies on top of part or all of a side of the pentagon.

- A) 4 B) 5 C) 6 D) 7 E) Answer not given.

7 Imagine all the ways you can draw a line, a square, and a circle of any size, either overlapping or not overlapping. Let N equal the number of points of intersection that can be created by drawing these three shapes. What is the sum of all possible values of N ? Note: the line does not lie on top of any of the sides of the square.

- A) 57 B) 60 C) 67 D) 78 E) Answer not given.

Continued on Next Page

	USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #8 THROUGH #10. A popular math game is called "ZIP, ZAP, ZORP" - maybe you have played it before! In the game, one person chooses a 2-digit whole number with no repeated digits, and another person tries to guess the number. After each guess, the person who chose the number will give one of the following clues: • ZIP: No digit is correct • ZAP: One digit is correct, and it is in the right place • ZORP: One digit is correct, but it is in the wrong place. • ZORP ZORP: Two digits are correct, but they are both in the wrong place. • ZAP ZAP: Correct number! Examples: Player A chooses 79 as the secret number. Player B guesses 42. Player A says 'ZIP'. OR if Player B guesses 91, Player A says 'ZORP'.
8	Biff picks 43 as his secret number. Eho's first guess is 40. What is Biff's response? A) ZIP B) ZAP C) ZORP D) ZORP ZORP E) Answer not given.
9	Based on the events in problem 8, how many reasonable 2nd guesses are available to Eho? Note: a reasonable 2nd guess is one that uses the information gained from the response to the 1st guess to narrow down the list of choices as much as possible. A) 15 B) 16 C) 17 D) 18 E) Answer not given.
10	Continuing the game described in problems 8 and 9, Eho makes a reasonable 2nd guess, to which Biff responds with ZIP. Assuming Eho uses perfect logic, what is the probability that he guesses the right number on his 3rd guess? A) 1/10 B) 1/9 C) 1/8 D) 1/7 E) Answer not given.

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7th/8th grade — April 25, 2026

Key

Team Multiple Choice Contest – Answer Key

7th/8th grade

Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.

Answer	
1	C
2	B
3	C
4	D
5	E (9)
6	C
7	D
8	B
9	B
10	D

"Math Is Cool" Masters — 2025-26
7th/8th grade — April 25, 2026

Final Score (out of 20)

Room #

School Name

Team #

Team Multiple Choice Contest - 15 minutes - ~20% of team score

You will have 15 minutes to answer 10 multiple choice questions as a team. This test is the only test where you will be penalized for incorrect responses. You will receive two points for a correct letter response, zero points for leaving it blank, and minus one point for an incorrect response. When you are prompted to begin, tear off the colored answer sheet, pass out a copy of the test to each team member, and begin testing. **ONLY a letter response should be listed as an answer on this answer sheet. Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.**

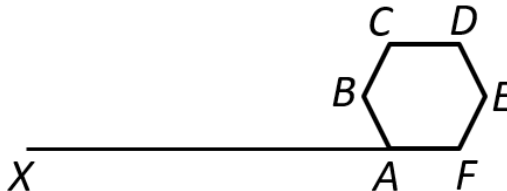
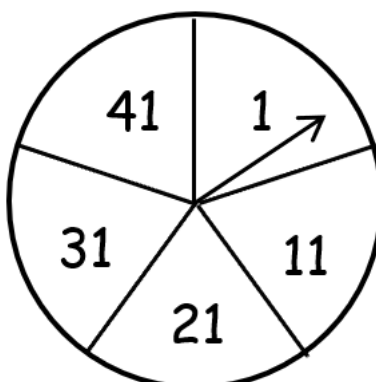
STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		-1, 0, or 2	-1, 0, or 2
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th grade		TOTAL:	

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Team Contest

1	What is the sum of all positive prime numbers less than 40 with at least one digit that is a 3?
2	Evaluate: $10000 \cdot (0.2)^4$
3	Find the sum $x + y$ of the numbers in the following sequence: 82, 68, x , 40, y , 12, -2, ...
4	A class of eight students had an average test score of 86. One student scored a perfect score of 100 and another received a 96. What was the mean test score of the other 6 students?
5	A tortoise walks at an average rate of 2 miles every 3 hours. A hare walks at an average rate of $\frac{2}{3}$ mile every 15 minutes. At these rates and in miles, how much farther will the hare walk in an hour than the tortoise?
6	Evaluate $678_9 - 456_7 - 234_5$ as a base-10 number.
7	A twelve-foot-long string, $\overline{AX}$, is attached to vertex A of regular hexagon $ABCDEF$, which has a perimeter of 12 feet. The string is kept taut (pulled straight), and wrapped around the hexagon, by moving in a circular clockwise direction. The area swept out by the string as it moves around the hexagon can be written as $R\pi/T$, where R and T are positive integers whose greatest common factor is 1. What is $R + T$? 
8	A circular spinner has 5 congruent sections, each labeled with one of the numbers, 1, 11, 21, 31, and 41. The spinner is spun twice, and the resulting numbers are recorded. As a reduced common fraction, the probability that the sum of the two numbers is divisible by 3 is A/B. What is $A + B$? 
9	An equilateral triangle and a regular hexagon have a combined perimeter of 66 cm and a combined area of $121\sqrt{3}$ cm ² . As a reduced common fraction, the side length of the hexagon is A/B cm. What is $A + B$?
Continued on next page.	

10 From a standard deck of playing cards, all 13 hearts have been lined up in a row, with positions 1 through 13, as indicated below. Face cards are defined as Jack, Queen and King. The Ace is the high card, so the natural order of the cards from left to right is 2 through 10, Jack, Queen, King, and Ace. Assume a value of 15 for the ace and face cards. Otherwise, assume their numerical value.

Given the following clues, determine the order of the 13 cards.

What is the sum of the card values in positions 5, 7 and 8?

Clues:

- No two cards are in their natural order from left to right.
- No face cards are at either end.
- Two face cards are adjacent.
- The King and Ace are separated by one card.
- The 2nd and 3rd cards have a sum of 13.
- The 10th and 11th cards have a sum of 10.
- The 8 is two places to the left of the ace.
- The Jack is two places to the right of the 2.
- The 4 is two places to the right of the 3.
- The 6 is three places to the left of the ace.
- The 9 is somewhere to the right of the 4.
- The 10 is two places to the right of the 5.
- The third card from the right and the third card from the left are both odd.

1st 2nd 3rd 4th 5th 6th 7th 8th 9th 10th 11th 12th 13th

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Key

Team Contest - Answer Key

7th/8th grade

Answer	
1	107
2	16
3	$[x + y =] 80$
4	82
5	2 [miles]
6	$251_{[10]}$
7	$[R + T =] 185$
8	$[A + B =] 34$
9	$A + B =] 49$
10	27

"Math Is Cool" Masters — 2025-26
7th/8th grade — April 25, 2026

Final Score (out of 10)

Room #

School Name

Team #

Team Contest - 15 minutes - ~30% of team score

When you are prompted to begin, tear off the colored answer sheet and give a copy of the test to each of your team members and begin testing. Each problem is scored as a 1 or 0. Record all answers on this colored answer sheet.

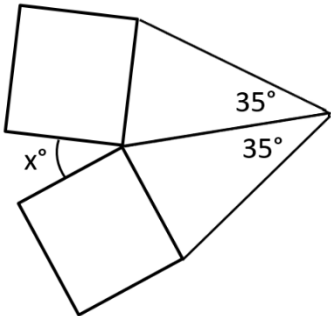
STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th grade		TOTAL:	

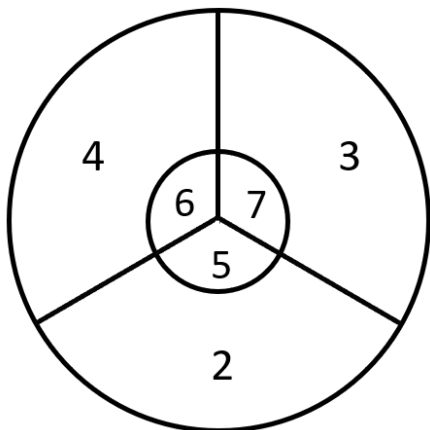
"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

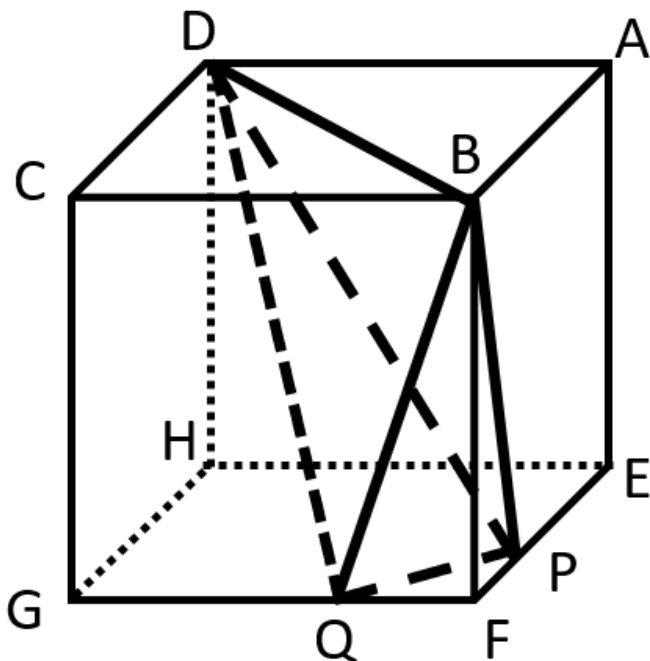
Linda Moore Triple Jump

1	Elodie is stacking identical chairs after a meeting. One chair by itself is 31 inches tall. A stack of 4 chairs is 40 inches tall. How tall, in inches, will a stack of 6 chairs be?																																																	
2	The 12 faces of a fair 12-sided die are labeled with the numbers 1 through 12. After the die is rolled once, as a simplified fraction, the probability that the number showing is a multiple of either 2, 3, or both is A/B . What is $A + B$?																																																	
3	What is 40% of 180% of 25?																																																	
4	The median of the set of integers shown is equal to the mode of the set. What is the value of x ? $\{18, 52, 26, 31, 47, x\}$																																																	
5	In the diagram, two congruent isosceles triangles share a side, and have one angle measuring 35°. Attached to each triangle is a rectangle that shares a side with the base of the triangle. In degrees, what is the measure of angle x? 																																																	
6	Ninety-five identical cubes are divided into two piles. All of the cubes in the first pile are placed 5 to a bag, and all of the cubes in the second pile are placed 8 to a bag. If a total of 13 bags are used, how many cubes are in the larger of the two piles?																																																	
7	In the following puzzle, place the digits 1 through 9 in the outlined empty squares, using each digit exactly once, in order to make the three horizontal and three vertical equations true. Calculations are done from left to right and top to bottom, NOT using the usual "order of operations". For example: $3 + 4 \times 2 = 7 \times 2 = 14$. What is the sum of the two numbers indicated by question marks? <table><tr><td>?</td><td>-</td><td></td><td>+</td><td></td><td>=</td><td>9</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td></td><td>+</td><td></td><td>$\times$</td><td></td><td>=</td><td>18</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>-</td><td></td><td></td></tr><tr><td></td><td>+</td><td>?</td><td>$\times$</td><td></td><td>=</td><td>26</td></tr><tr><td>=</td><td></td><td>=</td><td></td><td>=</td><td></td><td></td></tr><tr><td>17</td><td></td><td>15</td><td></td><td>3</td><td></td><td></td></tr></table>	?	-		+		=	9	+		+		-				+		$\times$		=	18	+		+		-				+	?	$\times$		=	26	=		=		=			17		15		3		
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17		15		3																																														

- 8 On the dartboard shown, the outer circle has a radius of 18 cm and the inner circle has a radius of 6 cm. Three radii divide each circle into three congruent regions, with point values shown. The probability that a dart will hit a given region is proportional to the area of the region. When two darts hit this board, the score is the sum of the point values of the regions hit. As a simplified fraction, the probability that the score is odd is A/B . What is $A + B$?



- 9 The cube with vertices A, B, C, D, E, F, G, H has edges of length 3 units. Point P is placed so that $FP/FE = 1/3$ and point Q is placed so that $FQ/FG = 1/3$. In cubic units, what is the volume of the tetrahedron with vertices B, D, P , and Q ?



- 10 Three numbers have a sum of 13 and the sum of their squares is 181. If the product of the three numbers is -78 , what is the largest of the three numbers?

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Key

Linda Moore Triple Jump - Answer Key

7th/8th grade

Answer	
1	46 [inches]
2	[A + B =] 5
3	18
4	[x =] 31
5	[x =] 35 [°]
6	80 [cubes]
7	11
8	[A + B =] 1069
9	3 [cubic units]
10	13

"Math Is Cool" Masters — 2025-26
7th/8th grade — April 25, 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

When you are prompted to begin, tear off the three colored answer sheets and give a copy of the test to each of your team members and begin testing. Record all answers on this colored answer sheet. This Submittal #1 will be collected after 5 minutes.

SUBMITTAL #1

STUDENTS: DO NOT WRITE IN SHADED REGIONS

		Scorer 2	Scorer 1
Answer		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7th/8th grade		TOTAL:	

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #2 will be collected after 10 minutes.

SUBMITTAL #2

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2 0 or 1	Scorer 1 0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th grade		TOTAL:	

"Math Is Cool" Masters — 2025-26
7th/8th grade — April 25, 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #3 will be collected after 15 minutes.

SUBMITTAL #3

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
7 th /8 th grade		TOTAL:	

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Proctor
Copy

Mental Math Contest

MENTAL MATH - 30 seconds per question - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

1	What is twenty-five times four minus twenty-five times two?	
2	Evaluate ten X minus X squared if X equals four.	
3	The lengths of the two legs of a right triangle are five and twelve inches. What is the number of inches in the perimeter of the triangle?	
4	Eho has one dollar and fifty-seven cents in his pocket. What is the largest number of nickels he could have in his pocket?	
5	There are five thousand two hundred and eighty feet in a mile. How many feet are in two-thirds of a mile?	
6	A sequence begins with the numbers three, six, ten, and fifteen. What is the eighth term in this sequence?	
7	What is the largest three-digit prime number whose digits add up to four?	
8	A standard deck of cards has all its face cards and aces removed. As a reduced common fraction, the probability that two cards, randomly selected without replacement, are both red and divisible by three is A over B. What is A plus B?	

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Key

COLLEGE BOWL ROUND #1

#	Problem	Answer
1	When X is divided by ten and then multiplied by five, the result is twenty. What is X ?	$[x =] 40$
2	A field has only chickens and sheep. There are seventy-four feet in the field. If there are twenty-one chickens, how many sheep are in the field?	8 [sheep]
3	The radius of a circle is thirteen point five centimeters. The circumference of the circle is A times π centimeters. What is A ?	$[A =] 27$
4	What is the smallest three-digit positive integer that is divisible by four, six, and nine?	108
5	A container in the shape of a rectangular prism holds two cubic feet of sand. The area of the base of the container is ninety-six square inches. In feet, what is the height of the container?	3 [feet]
6	Two standard six-sided dice are rolled. One is red and one is green. As a simplified fraction, the probability that the number on the green die is less than the number on the red die is A over B . What is A plus B ?	$[A + B =] 17$
7	Celine writes a list of ten consecutive integers. The sum of the least and greatest numbers on her list is ninety-three. What is the smallest number in her list?	42
8	The probability that a randomly chosen positive single-digit integer is divisible by two is A over B . What is A plus B ?	$[A + B =] 13$
9	If A star B equals two times A plus three times B , what is the value of four star two?	14
10	What is the mean of the numbers twenty-five, eight, seventeen, three, and seven?	12 [= mean]

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Key

COLLEGE BOWL ROUND #2

#	Problem	Answer
1	What is two hundred minus seventy-five plus fifty minus eighty?	95
2	The median of eight distinct positive integers is ten. What is the smallest possible integer in the set?	1
3	The first three terms of a geometric sequence are two, six, and eighteen. What is the fifth term in the sequence?	162 [= 5 th term]
4	How many ways are there to arrange the letters in the word BURRITO, spelled B - U - R - R - I - T - O, if the Rs must not be next to each other?	1800 [ways]
5	A rectangle has its length multiplied by one-point-five and its width multiplied by two-point-five to create a new rectangle. What percent of the area of the original rectangle is the area of the new rectangle?	375 [%]
6	Using the digits one through eight exactly once each, form two four-digits numbers with a sum as small as possible. What is the sum?	3825 [= sum]
7	A fish's head is six centimeters long. Its tail is the same length as the head plus two-thirds of the body. The body is as long as the head and tail together. In centimeters, how long is the fish?	72 [cm]
8	A rectangular prism has dimensions eleven inches by seventeen inches by nineteen inches. What is the number of cubic inches in the volume of the prism?	3553 [in ³]
9	Athena walks at an average rate of three miles per hour. How many minutes will it take her to walk zero-point-seven-five miles?	15 [minutes]
10	Solve the equation for X: thirty-two X minus eighty equals one hundred and forty-four	[x =] 7

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Key

COLLEGE BOWL ROUND #3

#	Problem	Answer
1	A jar has nine red, four green, and seven purple marbles in it. As a percentage, what is the probability that a randomly chosen marble is not green?	80 [%]
2	Evaluate $X^2 - Y^2$ if X equals nine and Y equals negative four.	65
3	How many square inches are in one-point-seven-five square feet?	252 [in^2]
4	A data set consists of thirteen numbers and has a mean of seventeen. What number, when added to the set, will cause the mean to increase to nineteen?	45
5	The area of the largest circle that can be inscribed in a regular hexagon is forty-eight pi square centimeters. What is the number of centimeters in the perimeter of the hexagon?	48 [cm]
6	The Fibonacci Sequence starts with zero, one, one, two, three, where, starting with term three, each term is the sum of the two previous terms. Let N be the number of terms in the sequence consisting of exactly three digits. What is N ?	[$N =$] 5
7	What is the greatest common factor of nine hundred sixty, and six hundred thirty?	30 [= gcf]
8	Point A with coordinates three comma two and point B with coordinates negative eight comma two are plotted on a coordinate plane. What is the number of units in the distance between point A and point B?	11 [units]
9	A drawer has two brown socks, four blue socks, three black socks, and six white socks. What is the minimum number of randomly chosen socks needed to ensure that three white socks are selected?	12 [socks]
10	What is forty-six times fifty-four?	2484

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Key

COLLEGE BOWL ROUND #4

#	Problem	Answer
1	The area of a triangle is sixty square inches. If the height of the triangle is fifteen inches, what is the number of inches in the base?	8 [inches]
2	What is the largest possible number of Wednesdays that can occur during the months of February and March of a given year?	9 [Wednesdays]
3	The first three terms of an arithmetic sequence are two, eleven, and twenty. What is the twentieth term in the sequence?	173 [= 20 th term]
4	Biff takes half an hour to trim the hedge. Eho takes twenty-five minutes to trim the same hedge. As a simplified fraction, the fraction of an hour it takes them to trim the hedge if they work together is A over B . What is A plus B ?	[$A + B =$] 27
5	The point with coordinates negative three comma four is on the graph of the line with equation Y equals $M X$ plus twenty-five. What is the value of M ?	[$M =$] 7
6	In a set of four distinct positive integers the range is twenty and the mean is thirteen. What is the largest possible median for this data set?	15
7	It takes nine accountants eight days to file thirteen tax returns. How many accountants would it take to file thirty-nine tax returns in three days?	72 [accountants]
8	What is the smallest odd composite number with exactly three distinct prime factors?	105
9	If N consecutive numbers are added, the positive difference between the largest and smallest of the consecutive numbers is N minus X . What is X ?	[$x =$] 1
10	What is fourteen-thirds divided by two-ninths?	21

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Key

COLLEGE BOWL ROUND #5

#	Problem	Answer
1	Let A equal twenty-two percent of one hundred and ten. Let B equal fifty-two percent of one hundred and ten. What is B minus A?	$[B - A =] 33$
2	How many centimeters are equivalent to nine hundred and fifty millimeters?	95 [cm]
3	What is six hundred and sixteen divided by eleven?	56
4	The first three terms in an arithmetic sequence are: twenty-one halves, eleven, and twenty-three halves. What is the 6 th term?	13 [= 6 th term]
5	Naveen has quarters and dimes in his pocket in a ratio of Q to D. If instead, the coins were in a ratio of D quarters to Q dimes, the value of his coins would be less by ninety cents. How many more quarters than dimes does he have in his pocket?	6 [more quarters]
6	What is the sum of the factors of eighty-four?	224
7	How many three-digit integers exist, such that their digits form an arithmetic sequence with a positive common difference? Note: the number three hundred twenty-one forms an arithmetic sequence with a negative common difference.	16 [integers]
8	X plus X plus six equals X plus X plus X plus X. What is the value of X plus two?	5 [= $x + 2$]
9	Josh and Grace both draw one card at random from a standard deck. Josh draws first and gets a red card. As a simplified fraction, the probability that Grace also draws a red card is A over B. What is A plus B?	$[A + B =] 76$
10	What is the median of the following set of numbers? Fifty, twenty, ninety, forty-four, eighteen, and sixty-two	47 [= median]

"Math Is Cool" Masters — 2025-26

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Key

COLLEGE BOWL ROUND #6

#	Problem	Answer
1	The mean of four numbers is twenty-seven. Three of the numbers are fourteen, thirty-five, and fifty-one. What is the fourth number?	8
2	How many positive two-digit integers have only odd digits?	25
3	There are five thousand two hundred and eighty feet in one mile. How many feet are in two point three miles?	12144 [feet]
4	The ratio of the areas of two similar rectangles is twelve to seventy-five. As a simplified fraction, the ratio of the rectangles' perimeters is A over B. What is A plus B?	$[A + B =] 7$
5	What is the smallest 3-digit sum of two prime numbers?	100
6	A field has only llamas and emus. There are a total of one hundred and thirty-four feet and forty heads in the field. How many more llamas are there in the field than emus?	14
7	Huey takes twenty-four minutes to read an article. Dewey can read the same article twice as fast as Huey. Louie can read the same article three-fourths as fast as Dewey. In minutes, how long does it take Louie to read the article?	16 [minutes]
8	Six apples cost the same as five pears. Four pears cost the same as eight oranges. How many oranges cost the same as three apples?	5 [oranges]
9	The first three terms of a geometric sequence are one hundred sixty-two, one hundred eight, and seventy-two. What is the fifth term in the sequence?	32
10	As a simplified fraction, the sum of one-third plus three-fifths plus seven-twelfths is A over B. What is A plus B?	$[A + B =] 151$

"Math Is Cool" Masters — 2025-26

7th/8th grade — April 25, 2026

Key

COLLEGE BOWL - EXTRA QUESTIONS

#	Problem	Answer
1	As a simplified fraction, the solution to the equation eleven X over twelve minus seven over eighteen equals five X over six plus thirteen over twenty-four is X equals A over B. What is A plus B?	[A + B =] 73
2	Beginning with the third term, a certain sequence grows by adding all previous terms together to get the next term. If term one and term two are one and four, then term N will be the first term greater than one hundred. What is N?	[N =] 8
3	What is the mean of the odd numbers between six and sixteen?	11
4	What is one thousand two hundred thirty-four minus three hundred forty-five?	889
5	The first and fifth terms of an arithmetic sequence are three-point-two and fifteen-point-two. What is the common difference?	3
6	An animal shelter has only cats and dogs. There are thirty-eight animals, twenty of which are cats. As a simplified fraction, the ratio of dogs to cats is A over B. What is A plus B?	[A + B =] 19
7	Leo is going to fill a box with measurements forty by forty by sixty units, using identical cubes. What is the minimum number of cubes needed?	12 [cubes]
8	Shanteel has read one third of her five-hundred-and-forty-page book at an average rate of twelve pages every fourteen minutes. At this rate, how long, in hours, will it take her to read the remaining pages?	7 [hours]
9	A coin purse contains only nickels, dimes, and quarters. The ratio of the number of nickels to dimes to quarters is ten to seven to three. If a quarter and a nickel are added, the total value of the coins in the purse becomes exactly twelve dollars. How many dimes are in the purse?	42 [dimes]
10	A fair twenty-sided die numbered one through twenty is rolled. As a percentage, what is the probability that the number showing is a single digit?	45 [%]